

12V Sodium-Ion Battery



12V

NOMINAL
VOLTAGE

30-100Ah

CAPACITY
RANGE

3K+

CYCLE
LIFE

95%

ROUND-TRIP
EFF.

-40/+60°C

OPERATING
TEMP

Zero

MAINTENANCE

10yr+

DESIGN
LIFE

IP65

ENCLOSURE
RATING

ENERGY STORAGE YOU CAN BANK ON

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Series S 12V Battery

- Sodium-Ion NFPP
- Drop-In Lead-Acid Replacement
- 30 / 60 / 100 Ah Models



The Series S 12V is a direct drop-in replacement for lead-acid batteries across mobility, off-grid, marine, and commercial applications. Built on Coulomb Technology's Sodium-Ion NFPP chemistry, it delivers **10× longer cycle life**, zero maintenance, and a 40% weight reduction — with no thermal runaway risk and no toxic materials.

Available in three capacity configurations:

- Series S 12V — 30Ah (360 Wh)
- Series S 12V — 100Ah (1.2 kWh)
- Standard BCI group form factors
- Series S 12V — 60Ah (720 Wh)
- Integrated BMS with cell balancing
- Bluetooth monitoring (optional)

KEY BENEFITS



Zero Thermal Runaway

Sodium-Ion NFPP chemistry is inherently non-flammable — no thermal runaway risk, no toxic gas emissions, and no special shipping restrictions.

- ✓ Air-shippable — no hazmat restrictions
- ✓ No hydrogen gas emissions
- ✓ Non-toxic, fully recyclable
- ✓ Simpler installation approvals



Extreme Temperature Range

Operates from -40°C to +60°C without derating — outperforming lead-acid in cold climates and high-heat environments alike.

- ✓ -40°C to +60°C discharge range
- ✓ No capacity loss in cold weather
- ✓ 3,000+ cycles at 80% DoD
- ✓ 10+ year design life



True Drop-In Replacement

Standard 12V form factor with BCI-compatible dimensions — replaces lead-acid with no system modifications, no new chargers, no rewiring.

- ✓ BCI group-compatible dimensions
- ✓ Works with existing chargers
- ✓ Zero maintenance — no watering
- ✓ 40% lighter than lead-acid

WHERE THE SERIES S PERFORMS



GOLF CARTS / SCOOTERS / LIGHT EV

Mobility & Light EV

Direct replacement for lead-acid in golf carts, electric scooters, and light electric vehicles — with 3× the range per charge and zero maintenance.



MARINE / RV / OFF-GRID

Marine & Off-Grid Power

Lightweight, vibration-resistant, and fully sealed for marine, RV, and off-grid solar applications — with no outgassing and no watering required.



WAREHOUSE / LOGISTICS

Material Handling

Fast charging and extreme temperature tolerance for 24/7 warehouse operations — pallet jacks, automated guided vehicles, and material handling equipment.



TELECOM / UPS / CRITICAL BACKUP

Backup & Telecom Power

Reliable backup power for telecom towers, UPS systems, and critical infrastructure — with 10+ year design life and zero maintenance overhead.

SERIES S — 12V BATTERY



NOMINAL VOLTAGE 12v 12.8V nominal (4S NFPP)	CAPACITY RANGE 30–100Ah Three models available	CYCLE LIFE 3K+ At 80% DoD, 25°C
WEIGHT (100AH) 13kg 40 % lighter than lead-acid	OPERATING TEMP –40/+60°C Discharge range	EFFICIENCY 95% Round-trip efficiency

PARAMETER	VALUE / DESCRIPTION	UNIT
NOMINAL VOLTAGE	12.8 (4S configuration)	VDC
CAPACITY — 30AH MODEL	30 (360 Wh)	Ah
CAPACITY — 60AH MODEL	60 (720 Wh)	Ah
CAPACITY — 100AH MODEL	100 (1,200 Wh)	Ah
USABLE CAPACITY	80–100% DoD (vs. 50% lead-acid)	—
CYCLE LIFE	3,000+ @ 80% DoD, 25°C	—
CELL TYPE	Sodium-Ion NFPP (Non-Flammable Polymer Pouch)	—
DISCHARGE TEMP RANGE	–40°C to +60°C	—
CHARGE TEMP RANGE	–10°C to +45°C	—
SELF-DISCHARGE RATE	<3% per month	—
ROUND-TRIP EFFICIENCY	~95%	—
BMS COMMUNICATION	Integrated BMS; optional Bluetooth monitoring	—
ENCLOSURE RATING	IP65 — sealed, vibration-resistant	—

CERTIFICATIONS UL 1973UL 9540UN 38.3 Cell & Module Level	UL 9540A TESTING Cell: Tested & Passed Module/System: In Progress	WARRANTY Cell: 3 Years System: 5 Years Extended warranty available
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QUALITY ASSURANCE & TEST SCOPE

- ✓ Visual, Mechanical & Dimensional Inspection

✓ 3 × 0.2C Capacity Verification (80% DoD)
- ✓ BMS Protections & Isolation Tests

✓ Baseline OCV & Self-Discharge Measurement
- ✓ Functional & Conditioning Cycles

✓ Vibration & IP65 Seal Verification

WHY SODIUM-ION NFPP OUTPERFORMS LEGACY LEAD-ACID TECHNOLOGY

A Clear Advantage Across Every Performance Dimension

Coulomb Technology's Sodium-Ion NFPP chemistry delivers superior cycle life, wider operating temperature range, zero maintenance, and a fundamentally safer chemistry — making it the ideal replacement for lead-acid in critical commercial and industrial applications.

FEATURE	SODIUM-ION NFPP	LEAD-ACID
Cycle Life	3,000+ cycles @ 80% DoD	300–500 cycles @ 50% DoD
Weight (100Ah)	13 kg (28.7 lb)	~22 kg (48.5 lb)
Usable Capacity	80–100% DoD	50% DoD (to preserve life)
Discharge Temp Range	–40°C to +60°C	–20°C to +50°C
Charge Temp Range	–10°C to +45°C	0°C to +40°C
Maintenance	Zero maintenance	Regular watering & equalization
Self-Discharge	<3% per month	5–15% per month
Charging Efficiency	~95%	~75%
Thermal Runaway Risk	Zero risk	Low risk
Toxic Gas Emissions	None	Hydrogen gas during charging
Memory Effect	None	Sulfation if undercharged
Lifespan	10+ years	3–5 years
Environmental Impact	Non-toxic, recyclable	Lead & acid hazardous

30%

Safer chemistry
vs. LFP lithium-ion

50%

More extreme temp
tolerant vs. LFP

20%

Less degradation
vs. LFP lithium-ion

Zero

Thermal runaway
risk

SYSTEM ADVANTAGE OVER LEAD-ACID

Lower Total Cost of Ownership Over the Product Lifetime

Sodium-Ion NFPP eliminates replacement costs, maintenance overhead, and hazardous disposal — delivering measurable savings vs. lead-acid through zero maintenance, longer design life, and higher efficiency.

CYCLE LIFE

3,000+

3,000+ cycles at 80% DoD vs. 300–500 cycles for lead-acid — dramatically fewer replacements over the product lifetime.

DESIGN LIFE

10+ Years

Sodium-ion NFPP chemistry delivers a 10+ year design life vs. 3–5 years for lead-acid — reducing replacement and disposal costs.

ROUND-TRIP EFFICIENCY

≥95%

~95% round-trip efficiency vs. ~75% for lead-acid — more usable energy from every charge cycle.

ZERO MAINTENANCE

Zero

No watering, equalization, or scheduled maintenance — eliminating 100% of lead-acid O&M overhead.

WEIGHT SAVINGS

40%

40% lighter per 100Ah vs. lead-acid — reducing freight, handling, and installation labor costs.

HAZARDOUS DISPOSAL

None

Non-toxic, fully recyclable chemistry eliminates hazardous lead & acid disposal fees and regulatory burden.

RESEARCH & DEVELOPMENT PARTNERS

